

A certified three-way triage gate for industrial visual inspection: escaped-defect and false-reject guarantees on MVTec AD, VisA, and MPDD

Zeyu Fu*

July 17, 2026

Abstract

Anomaly detectors reach near-ceiling image-AUROC on MVTec AD, yet AUROC is not a deployable decision: it names no operating threshold and bounds neither asymmetric-cost error — passing a defective part (an *escaped defect*) or rejecting a good one (a *false reject*). We present *inspect-gate*, a distribution-free triage layer sending each image to auto-pass, auto-reject, or defer on any backbone’s scores, with per-category certified escaped-defect ($P(\text{auto-pass} \mid \text{defective}) \leq \alpha_{\text{miss}}$) and false-reject ($P(\text{auto-reject} \mid \text{good}) \leq \alpha_{\text{fr}}$) rates. Both reduce exactly to split-conformal calibration — false-reject directly, escaped-defect through a score-negation order-statistic identity — adding no new estimator, inheriting finite-sample validity. A category whose calibration pool is too small is *refused* and reported audited-not-certified, never rounded. The second certificate’s payoff: against the nearest single-threshold alternative (conformal risk control), it holds a lower false-reject rate at equal escaped-defect on VisA (3.0% vs. 16.2% at 16.4% deferral) and MVTec AD; on MPDD the contrast is starkest (36.9% $\rightarrow$ 0.0%) but only because the gate auto-rejects nothing there, deferring 73.1% instead. No kill-gate tripped across two backbones and five seeds. On MVTec AD, escaped-defect certifies in all 15 categories, false-reject only in the four clearing the good-calibration floor (4/15, a pool-size artifact) — a preregistered train-holdout remedy (PatchCore only) lifts the deployment-relevant figure to 12–13 of 15. On VisA both axes certify in all 12 categories and the deferral-skill audit becomes informative. MPDD (painted-metal parts) is the stingy extreme: its uniformly small good pools certify false-reject in 0/6 categories (escaped-defect still 5/6), and a flag-gated train-holdout arm rescues 4/6 — refusal tracking the data throughout (0/6 $\rightarrow$ 4/15 $\rightarrow$ 12/12).

1 Introduction

Automated visual inspection is one of the most mature industrial applications of computer vision, and unsupervised anomaly detection (AD) on the MVTec AD benchmark (Bergmann et al., 2019) is its standard proving ground. Modern detectors — memory-bank methods such as PatchCore (Roth et al., 2022) and reconstruction/transformer methods such as Dinomaly (Guo et al., 2025) — report image-level AUROC above 0.98 and, for the newest models, above 0.99. On the usual reading the problem looks solved.

*State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Army Medical University, Chongqing, China. ORCID: 0009-0001-8329-0108. Corresponding author: fuzeyu09@gmail.com.

It is not solved as a decision. Image-AUROC is a threshold-free ranking summary: it says a detector orders defective above good images well, but it names no operating point, and on a real line an operating point is exactly what is needed. Worse, the two mistakes a threshold can make are not symmetric. Passing a defective part (an *escaped defect*) can ship a fault to a customer; rejecting a good part (a *false reject*) wastes yield and, at high rates, risks eroding operator trust in the system. A useful inspection system must bound both, per product category, with a guarantee that holds in finite samples rather than on average over an idealized distribution.

We take the position that the deployable object is not a better detector but a certified *triage layer* on top of an existing one. Concretely, we route each image into three actions: *auto-pass* (confident-good, ship without review), *auto-reject* (confident-defective, scrap or rework without review), and *defer* (send to a human). The value proposition is that the two automated actions carry certificates: the auto-pass region is calibrated so that the probability a truly defective image lands there is at most α_{miss} , and the auto-reject region so that the probability a truly good image lands there is at most α_{fr} ; everything else defers. The human-review budget is then whatever the data forces, and — crucially — it is visible: a category whose data cannot support the requested guarantee defers more, rather than pretending to a certificate it cannot honor. We evaluate the layer on three complementary benchmarks by design: MVTec AD, whose small per-category good pools stress the refusal machinery; VisA, whose lower score ceilings and larger good pools let both certificates bind everywhere and give the deferral-skill audit real headroom; and MPDD (real painted-metal-parts fabrication), whose uniformly small good pools make it the stingiest certifiability point — false-reject certifiable in 0/6 categories under the primary protocol, the extreme that shows the gate refuses precisely when the data cannot pay for a certificate.

This framing turns a benchmark score into an auditable decision and makes three contributions.

- **A certified three-way triage system for inspection.** The contribution is not a new estimator — both guarantees reduce exactly to split-conformal quantiles (Section 3.1: false-reject directly, escaped-defect through a score-negation order-statistic identity) — but the coupled system assembled from them: two simultaneously certified error axes (escaped-defect and false-reject), explicit per-category certifiability floors, first-class refusal, and the K1–K7 kill-gate battery, packaged as an auditable decision layer over any backbone’s scores. To our knowledge no published method delivers this exact combination on MVTec AD or VisA; the binding K6 pre-submission citation re-scan, executed after the freeze (Section 2), returned a clear verdict — no scoop. Section 5.4 makes the coupled system’s added value concrete rather than asserted: recomputed on the identical score dumps, the single-threshold construction our own escaped-defect certificate specializes (conformal risk control) pays 36.9% false-reject on MPDD to hold the same escaped-defect rate our dual gate holds at 0.0%, purely by auto-rejecting the ambiguous band instead of deferring it. We foreground the “no new math” property deliberately: it is what makes the guarantees finite-sample-exact and the tool a thin, testable wrapper rather than a heuristic.
- **Refusal as a first-class outcome, with teeth.** A per-category certifiability floor $\alpha_{\text{min}} = 1/(n_{\text{cal}} + 1)$ makes “we cannot certify this category at this rate” an explicit, reported state (audited-not-certified), never a silently loosened threshold. This is not cosmetic: on our own substrate (Table 3), a non-refusing baseline that issues an auto-reject threshold and claims the requested false-reject rate $\alpha_{\text{fr}} = 0.05$ in the 11 categories the gate refuses would be asserting a guarantee the calibration pool cannot support — the achievable floor there is 0.06–0.14 (e.g. toothbrush $1/7 = 0.14$), $1.2\times$ – $2.9\times$ looser than claimed — so it silently over-promises where the gate refuses and defers. (On the escaped-defect axis the floor clears $\alpha_{\text{miss}} = 0.10$ in all

15 categories, so this failure mode does not arise there; a post-hoc binding demonstration (Section 5.11) confirms this empirically — in a range of (backbone, category) cells a non-refusing fixed-threshold baseline’s *realized* escaped-defect or false-reject rate exceeds the target while the certified gate holds within it on the same data.)

- **A practitioner-facing audit.** Beyond the certificate, we test whether field-standard threshold practice earns any deferral skill over honest random deferral (an excess-AURC audit), reporting a constructive or a debunking verdict symmetrically; the preregistered confirmatory pooled audit returns the constructive verdict — practice earns real deferral skill in all five seeds (Section 5.10).

Figure 1 summarizes the resulting pipeline, from a frozen backbone’s per-image score through the G1/G2 certificates and certifiability floor to the three-way auto-pass/defer/ auto-reject route.

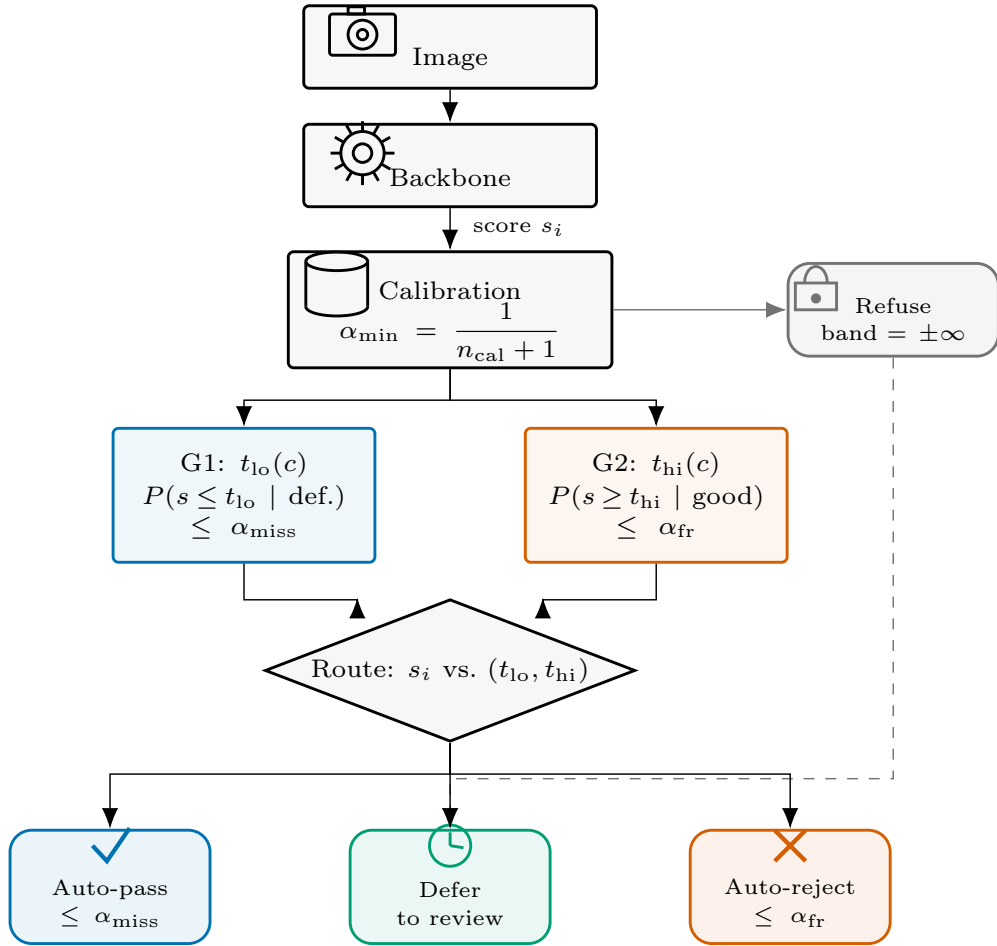


Figure 1: Method overview. A frozen backbone (PatchCore or Dinomaly, interchangeable) scores each image; per-category calibration checks the certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}} + 1)$ (Section 3.2) and, when met, sets the G1 escaped-defect threshold $t_{\text{lo}}(c)$ and the G2 false-reject threshold $t_{\text{hi}}(c)$ (Section 3.1). The three-way route then compares each score against $(t_{\text{lo}}, t_{\text{hi}})$ to auto-pass, defer, or auto-reject, carrying the stated escaped-defect and false-reject guarantees; an unmet floor instead refuses the axis (band = $\pm\infty$), defers, and is reported audited-not-certified.

We are explicit about status. The preregistration was frozen on 2026-07-11 (sha256-pinned sign-off; Section 6); this manuscript reports the primary-protocol axes on all three benchmarks plus the frozen G2 train-holdout promotion arm and two *post-freeze* benchmarks (VisA and MPDD — neither part of the frozen preregistration; both run under the identical frozen protocol and code path, their audits reported as exploratory and never in the confirmatory Holm family), and marks every remaining preregistered-but-uncomputed arm as an explicit slot. No uncomputed result is presumed anywhere in the text.

2 Related work

Distribution-free risk control. The gate is an application of conformal prediction (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023): given an exchangeable calibration set, an order statistic of the calibration scores yields a threshold with a finite-sample coverage guarantee. The broader risk-controlling line — risk-controlling prediction sets (Bates et al., 2021) and Learn-then-Test (Angelopoulos et al., 2021) — generalizes this to bounding arbitrary monotone risks and to multiple-hypothesis calibration. Our escaped-defect and false-reject bounds are the two one-sided special cases that split conformal already delivers exactly (Section 3.1); we deliberately use the simplest sufficient construction rather than the more general RCPS/LTT machinery, and cite the latter as the family the method belongs to.

Industrial anomaly detection. MVTec AD (Bergmann et al., 2019) defined the per-category unsupervised setting we use. PatchCore (Roth et al., 2022) (a coreset memory bank over pretrained features) and Dinomaly (Guo et al., 2025) (a transformer reconstruction model) are the two reproduced backbones here; EfficientAD (Batzner et al., 2024) is a fast alternative we discuss but do not certify (its only public implementations are community reimplementations, so it cannot anchor a binding reproduction gate). Dinomaly descends from the multi-class (one model, many categories) line initiated by UniAD (You et al., 2022) — the setting its published VisA figure reports and the one we use on VisA (Section 4). MPDD (Ježek et al., 2021) adds a real painted-metal-parts fabrication benchmark whose non-homogeneous backgrounds and small good pools make it a deliberately harsher industrial-realism stress test than the two academic staples — though, like them, still an academic benchmark rather than production-line data (see Limitations, Section 6). We use anomalib (Akçay et al., 2022) for the PatchCore implementation. Our contribution is orthogonal to all of these: we consume their scores and add a decision layer, and we improve none of them.

Selective prediction and abstention. The defer action is a selective-prediction mechanism (Geifman and El-Yaniv, 2017), but applied at the decision layer with a two-sided, per-category certificate rather than a single accuracy-coverage curve; the auto-reject certificate (false-reject control) has no analogue in standard selective classification, which abstains only around the positive decision.

Conformal anomaly detection and abstention. Conformal methods have been applied to anomaly detection before, but not to the certified inspection-triage decision we target. Early work bounded false-alarm rates for sequential and trajectory anomalies (Laxhammar and Falkman, 2011, 2015); recent tooling controls the false-discovery rate over flagged anomalies (Hennhöfer et al., 2026); conformal risk control (Angelopoulos et al., 2024) and its graph instantiation CRC-SGAD (Bai et al., 2025) bound a monotone risk (e.g. FNR/FPR) but emit a single flag, not a three-way route; and

Table 1: Notation and short-code glossary. Every code below recurs at least twice in Sections 3–6; consult this table rather than re-deriving a code’s meaning from context on each reuse.

Code	Meaning
G1	Certified escaped-defect bound (Section 3.1)
G2	Certified false-reject bound (Section 3.1)
K1–K7	Kill-gates: coverage sanity, vacuity, backbone floor (≥ 0.90 AUROC), audit headroom, calibration floor, prior-work re-scan, compute guard (in order)
V1 tier-1	Validity audit: certificate-validity statistic
V1 tier-2	Validity audit: stringent Clopper–Pearson estimator-variance readout
A1–A3	Amendments: (A1) tier-2 power floor graded per repeat, not pooled; (A2) near-target tier-2 estimator variance is expected, not failure; (A3) false-reject axis graded only over G2-certified cells
B1/B2/B3	Threshold practices: (B1) single global fixed threshold; (B2) per-category tuned threshold; (B3) train-good quantile threshold
C2	Confirmatory excess-AURC audit (pooled Holm family, Section 5.10)
CRC	Conformal Risk Control, the single-threshold conformal baseline

dual-threshold conformal abstention for autonomous-system perception (Kumar et al., 2025) adds a reject option on camera/LiDAR (and classification) tasks, with no industrial-inspection estimand and no evaluation on MVTec AD. The closest defect-detection application (Shen and Liu, 2025) bounds a single mean error rate for surface-defect segmentation, again with no deferral option and not on MVTec AD. To our knowledge no published method delivers a certified *three-way* triage (auto-pass / auto-reject / defer) on MVTec AD with *coupled* escaped-defect and false-reject guarantees and explicit per-category certifiability floors; the preregistered K6 scoop gate, run as a fresh pre-submission re-scan after the freeze, confirms this — the closest published systems (Kumar et al., 2025; Bai et al., 2025; Shen and Liu, 2025) each differ on a load-bearing axis (substrate, estimand coupling, or the defer option), and two adjacent 2026 works (a selective-risk certificate on generic vision, and a conformal cyber-physical-systems detector on sensor time-series) target neither the image-inspection estimand nor the three-way route. That is the seam this paper occupies. Section 5.4 instantiates the single-threshold CRC construction (Angelopoulos et al., 2024) as a quantitative baseline on our own canonical score dumps under the identical protocol, isolating what the three-way route and its coupled false-reject certificate add over the nearest single-flag alternative. More broadly, a recurring reliability finding is that low aggregate error coexists with brittle or miscalibrated decisions; our escaped-defect vs. false-reject framing is the industrial-inspection instance of that gap, and the excess-AURC audit (Section 3.4) tests whether standard thresholding beats chance at spending a deferral budget.

3 Methods

Table 1 collects the short-codes reused throughout Sections 3–6, as a single reference point instead of a re-derivation on each reuse.

3.1 The three-way gate and its two certificates

Each image i in category c has an anomaly score s_i with the convention *higher = more anomalous*. The gate is a pair of per-category thresholds $(t_{\text{lo}}(c), t_{\text{hi}}(c))$ with $t_{\text{lo}} \leq t_{\text{hi}}$: images with $s_i \leq t_{\text{lo}}(c)$ auto-pass, images with $s_i \geq t_{\text{hi}}(c)$ auto-reject, and the rest defer. (If a refusal or crossed thresholds leave the auto-reject band empty, the overlap simply defers, and both guarantees still hold.)

G2 — certified false-reject, directly. $t_{\text{hi}}(c)$ is the split-conformal upper quantile of the *good* calibration scores at level α_{fr} : with $n_{\text{cal}}^{\text{good}}$ good calibration scores, it is the order statistic at rank $\lceil (n+1)(1-\alpha_{\text{fr}}) \rceil$, guaranteeing $P(s \geq t_{\text{hi}} \mid \text{good}) \leq \alpha_{\text{fr}}$ under exchangeability. This is exactly the split-conformal upper-quantile construction applied to the good scores.

G1 — certified escaped-defect, by negation. Bounding $P(s \leq t_{\text{lo}} \mid \text{defective})$ is the mirror image: apply the same split-conformal construction to the *negated* defective calibration scores $Y = -X$ at level α_{miss} , then map the resulting upper quantile back through the negation. The gate module proves this order-statistic identity, and it is checked against the design’s $\lfloor \alpha(n+1) \rfloor$ formula in the accompanying unit tests. No new estimator is introduced; both certificates are the same conformal quantile.

Mondrian per-category stratification. Calibration is stratified by category (a Mondrian conformal construction): each category gets its own $(t_{\text{lo}}, t_{\text{hi}})$, so the guarantee is conditional on category, not marginal. An optional finer per-defect-type stratification is exploratory only (Section 6).

Where the labeled defectives come from. G1 calibrates on labeled *defective* images per category ($n_{\text{cal}}^{\text{def}}$ in Table 3) — a requirement unsupervised anomaly detection is deployed specifically to avoid needing at detector-training time, since labeled defects are scarce, especially early in a line’s life. This is a real cost relative to the unsupervised premise: the certificate is only as good as the labeled defective pool behind it, and unlike the detector itself that pool cannot be assembled from good parts alone. In deployment it is not a fixed benchmark quantity but accrues over time, from rework, scrap, and quality-escape logs as the line runs. The $\alpha_{\text{min}} = 1/(n_{\text{cal}}^{\text{def}} + 1)$ floor and the refusal rule below are what make this honest rather than aspirational: a category with a thin defect history is refused and reported audited-not-certified rather than issued a certificate its data cannot support, and MPDD’s connector category already demonstrates the mechanism in practice ($n_{\text{cal}}^{\text{def}} = 7 < 9$ at $\alpha_{\text{miss}} = 0.10$, refused at the frozen rate; Section 5.3).

3.2 Certifiability floors and refusal

Split conformal cannot certify an arbitrarily small rate from a finite calibration pool: the smallest achievable level is $\alpha_{\text{min}} = 1/(n_{\text{cal}} + 1)$. A category is certifiable for a guarantee iff $\alpha_{\text{min}} \leq \alpha$; otherwise the gate *refuses*, setting the corresponding threshold to $\pm\infty$ (the band is empty, all such images defer) and marking the category audited-not-certified for that axis. Refusal is emitted, never rounded away — this is the honesty rule that makes the floor table (Section 5.3) a result rather than a limitation.

3.3 Validity audit (V1) and kill-gates (K1–K7)

We fix $\alpha_{\text{miss}} = 0.10$ and $\alpha_{\text{fr}} = 0.05$ throughout, with a +3pp estimator tolerance (thresholds 0.13 / 0.08) and 95% one-sided Clopper–Pearson intervals (Clopper and Pearson, 1934). The *validity audit* V1 has two tiers. Tier-1 checks that the realized mean escaped-defect and false-reject rates lie within target+3pp over repeated calibration/evaluation splits; this is the certificate-validity statistic. Tier-2 forms a Clopper–Pearson upper bound and is graded, per the frozen amendments, as a stringent estimator-variance readout (Section 5.12), because on MVTec AD its power floor is not met per repeat and its tolerance is near-unpassable by construction for a correctly-calibrated gate.

A battery of preregistered kill-gates guards the construction: K1 (coverage sanity — halt if tier-1 fails in ≥ 5 of the $15 \times B$ cells), K2 (vacuity — kill if median deferral > 0.80 in $\geq 8/15$ categories on all backbones), K3 (backbone floor — mean image-AUROC ≥ 0.90), K4 (audit headroom), K5 (calibration floor — fires only if $\alpha_{\min} > 0.10$ in ≥ 3 categories), K6 (a citation re-scan for a prior certified triage), and K7 (a compute guard).

3.4 Excess-AURC audit

Beyond validity, we ask whether standard thresholding practice earns deferral skill at all. For each practice — B1 (global fixed threshold), B2 (per-category tuned), B3 (train-good quantile) — we compare its area-under-the-risk-coverage-curve against the closed-form random-deferral null (evaluated in closed form, never Monte-Carlo’d), using a matched-abstention, category-stratified permutation p -value (2000 permutations), a category-blocked bootstrap CI, and a roster-derived Holm (Holm, 1979) correction. The verdict is symmetric: if no practice beats random the paper reports a debunking finding; if all do it reports a constructive one.

4 Experimental setup

Data. Three public benchmarks. MVTec AD (Bergmann et al., 2019), staged from a fixed tarball (`sha256=cf4313b1...98cd3d`): 15 object/texture categories, 3,629 train-good images, and a test set of 467 good + 1,258 defective = 1,725 images across 73 defect types. VisA (Zou et al., 2022), staged from the official `VisA_20220922` archive under the official `spot-diff` 1-class split: 12 categories, 8,659 train-good images, and a test set of 962 good + 1,200 anomalous = 2,162 images (VisA’s public ground truth carries a single “bad” stratum per category, so the defect-type Mondrian axis is unavailable there). MPDD (Ježek et al., 2021) (post-freeze exploratory; see below), 6 real painted-metal-part categories in native MVTec-AD directory layout, staged from the credential-free public HuggingFace mirror `meksamiao/mpdd` (`archive sha256=69f8da73...ddd6`, recorded in the frozen split manifest): 888 train-good images and a test set of 176 good + 282 defective = 458 images. Both box layouts (MVTec AD, VisA) were built by symlinking the official split file, refusing on any count mismatch; for MPDD, which ships without an external split file, the ground-truth test split is a frozen per-category manifest generated from the verified staged tree (the role the official one-class split file plays for VisA). The score dumps were adapted to the canonical schema with a per-image join against that split that refuses on any missing or extra image.

Backbones (Branch A, $B = 2$). PatchCore (Roth et al., 2022) via anomalib (Akçay et al., 2022) (a Wide ResNet-50-2 backbone, features from the second and third stages, coreset ratio 0.10)¹ and Dinomaly (Guo et al., 2025) (canonical score dump, its own eval script; on VisA, the reference repo’s multi-class setting — one model per seed for all 12 categories — which is the setting its published VisA figure reports). Both were confirmed present and reproduction-passing before any calibration, so the preregistered two-backbone branch (Branch A) is active; the same two backbones and configurations are used unchanged on VisA.

¹The reproduction target 0.991 is PatchCore’s published *25%-coreset* mean image-AUROC (the paper’s own headline is “up to 99.6%”); our scoring uses the 10%-coreset configuration (PREREG D3), a different published operating point in the same paper. The frozen ± 0.02 tolerance spans the gap and the gate passes with margin either way (Table 2).

Table 2: Reproduction gate, 5 seeds per backbone per benchmark. Mean image-AUROC range is over seeds; the weakest category is named. “n/a”: no repo-confirmed published PatchCore-on-VisA/MPDD figure exists, so that row is descriptive by design (no pass/fail graded). Recomputed from the frozen MVTec AD, VisA, and MPDD analysis summaries.

Benchmark	Backbone	Target (± 0.02)	Mean I-AUROC (seed range)	Weakest category	Gate
MVTec AD	PatchCore	0.991	0.9817–0.9826	toothbrush 0.911–0.917	PASS 5/5
MVTec AD	Dinomaly	0.996	0.9957–0.9965	capsule 0.976–0.982	PASS 5/5
VisA	PatchCore	n/a	0.9033–0.9083	macaroni2 0.656–0.665	descriptive
VisA	Dinomaly	0.987	0.9866–0.9876	macaroni2 0.956–0.966	PASS 5/5
MPDD	PatchCore	n/a	0.9481–0.9568	bracket black 0.872–0.888	descriptive
MPDD	Dinomaly	0.972	0.9712–0.9738	bracket black 0.929–0.940	PASS 5/5

Protocol. Primary protocol: test-side good calibration, no train-holdout. For each (backbone, category, seed) we draw $R = 20$ stratified 50/50 calibration/evaluation splits (split-seed = repeat index), with 5 backbone seeds (0–4).

5 Results

All numbers in this section are recomputed from the frozen local score dumps by the archived analysis pipeline — a MVTec AD analysis stage, a stage-identical VisA analysis stage, a stage-identical MPDD analysis stage (with its own train-holdout rescue stage), and the G2-promotion stage, each emitting a frozen result summary; none is trusted from a training log or the literature.

5.1 Reproduction gate

On all three benchmarks the gate recomputes image-AUROC (Mann–Whitney rank-sum identity) directly from the local scores and compares it against a named published target (Table 2). On MVTec AD both backbones pass in all 5 seeds, and K3 (mean AUROC ≥ 0.90) holds with wide margin. On VisA, Dinomaly passes in all 5 seeds against its own published VisA figure (0.987, multi-class setting), and our recomputation again reproduces its reported per-seed mean image-AUROC seed for seed (max per-category deviation 3.3×10^{-5} over all 60 seed $\times$ category cells). PatchCore has no repo-confirmed published VisA figure, so per the tool’s no-guessed-target rule its VisA row is *descriptive* (no pass/fail is graded); its recomputed mean of 0.903–0.908 still clears the K3 floor, though with thin margin (seed 3: 0.9033, i.e. 0.33pp above the 0.90 floor), and the drop from 0.982 (MVTec) is exactly the lower-ceiling substrate the audit needs (Section 5.9). On MPDD, Dinomaly passes in all 5 seeds against its published MPDD multi-class figure (0.972; recomputed mean 0.9712–0.9738, grand mean 0.9722 ± 0.0010), our recomputation matching the box’s per-category training-log AUROC to $\leq 4.9 \times 10^{-5}$; PatchCore-on-MPDD has no repo-confirmed published figure, so per the same no-guessed-target rule its row is descriptive (recomputed mean 0.948–0.957).

Table 3: Per-category certifiability at $\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$ (backbone-invariant; pure functions of the shared MVTec test split). G1 certifies in 15/15, G2 in 4/15. Source: the frozen preregistration (§3), reproduced with 0/15 mismatch by the analysis pipeline.

Category	$n_{\text{cal}}^{\text{def}}$	$\alpha_{\text{min}}(\text{G1})$	G1@0.10	$n_{\text{cal}}^{\text{good}}$	G2@0.05
bottle	32	0.0303	OK	10	REFUSE
cable	46	0.0213	OK	29	OK
capsule	54	0.0182	OK	12	REFUSE
carpet	44	0.0222	OK	14	REFUSE
grid	28	0.0345	OK	10	REFUSE
hazelnut	35	0.0278	OK	20	OK
leather	46	0.0213	OK	16	REFUSE
metal_nut	46	0.0213	OK	11	REFUSE
pill	70	0.0141	OK	13	REFUSE
screw	60	0.0164	OK	20	OK
tile	42	0.0233	OK	16	REFUSE
toothbrush	15	0.0625	OK	6	REFUSE
transistor	20	0.0476	OK	30	OK
wood	30	0.0323	OK	10	REFUSE
zipper	60	0.0164	OK	16	REFUSE
Certifiable			15/15		4/15

5.2 Cross-seed stability

PatchCore’s five seeds are genuinely distinct: all 60 seed-0-vs-seed- N score-table comparisons (15 categories $\times$ 4 seeds) are 0/60 bit-identical, with $\max |\Delta \text{score}|$ ranging 0.026–0.153 (mean 0.074) — a real but modest variance source, so PatchCore is reported as a 5-effective-seed backbone, not collapsed to one. Dinomaly is low-variance: mean image-AUROC across seeds 0.9960 ± 0.00029 , with the widest per-category spread only 0.0060 (capsule). The same picture holds on VisA: PatchCore is 0/48 bit-identical ($\max |\Delta \text{score}|$ 0.044–0.256), and Dinomaly’s mean image-AUROC across seeds is 0.9870 ± 0.0004 with the widest per-category spread 0.0104 (macaroni2). On MPDD the pattern repeats once more: PatchCore is 0/24 bit-identical across the seed-0-vs-seed- N comparisons, and Dinomaly’s mean image-AUROC across seeds is 0.9722 ± 0.0010 .

5.3 Certifiability floors and the G1/G2 split

Table 3 is the paper’s honesty figure. The per-category calibration counts computed from the full local test arrays match the preregistered arithmetic-only floor table exactly (0/15 mismatches for both backbones, identical across all 5 seeds). At the frozen rates, escaped-defect (G1) is certifiable in all 15 categories; false-reject (G2) is certifiable in exactly 4 — cable, hazelnut, screw, transistor — the categories with $n_{\text{cal}}^{\text{good}} \geq 19$. The other 11 categories are refused for G2 and reported audited-not-certified. We are careful about what this exact match does and does not show: the per-category counts are a *deterministic* function of the fixed MVTec test split (round-half-to-even of $n/2$), so the 0/15 agreement with the preregistered table is a pipeline-correctness check — it confirms the data staging re-derives the frozen counts with no off-by-one or CSV-read error — not an empirical result about the gate. The genuinely empirical result is the V1 tier-1 validity audit below (Section 5.8), measured from real calibrate/route/evaluate runs, which could have failed had the calibration or routing pipeline been broken.

VisA: the same arithmetic, a friendlier split. On VisA the identical floor computation lands on the opposite side of the ledger: every category’s test split carries 100 anomalous and 50–101 good images, so after the 50/50 calibration split $n_{\text{cal}}^{\text{def}} = 50$ everywhere ($\alpha_{\text{min}}^{\text{G1}} = 0.0196$) and $n_{\text{cal}}^{\text{good}} \in \{25, 30, 50\}$ ($\alpha_{\text{min}}^{\text{G2}} = 0.0196\text{--}0.0385$), all below $\alpha_{\text{fr}} = 0.05$. G1 and G2 both certify in 12/12 categories under the primary protocol alone — no train-holdout remedy is needed on VisA — and the floors are again identical across both backbones and all five seeds (0 mismatches). This is the refusal mechanism working in both directions: it refuses 11/15 categories on MVTec AD because the data cannot support the rate, and certifies 12/12 on VisA because it can.

MPDD: the stingy extreme of pool-size-driven refusal. MPDD’s per-category test-good pools are uniformly small (26–32 images $\rightarrow n_{\text{cal}}^{\text{good}}$ 13–16, every one below the 19 the false-reject floor needs at $\alpha_{\text{fr}} = 0.05$), so G2 is certifiable in 0/6 categories under the primary protocol, on both backbones, identical across all five seeds (0 mismatches). G1 (escaped-defect) stays healthy at 5/6 — only connector fails, its 14 test-defective yielding $n_{\text{cal}}^{\text{def}} = 7 < 9$. This places MPDD at the stingy end of three points consistent with pool-size-driven refusal, ordered by good-pool size: MPDD 0/6 $\rightarrow$ MVTec AD 4/15 $\rightarrow$ VisA 12/12 (all three computed apples-to-apples by the identical floor code path). MPDD is thus not an intermediate point but the cleanest demonstration of the paper’s thesis — the certifiable count is a property of how much exchangeable good data the protocol can spend, and the gate refuses precisely when the data cannot pay for a 5%-false-reject certificate. Figure 2 makes this certify-or-refuse pattern visible at the category level across all three benchmarks: VisA certifies both axes in every category, MVTec AD’s false-reject row is hatched (refused) everywhere except the four categories in Table 3, and MPDD’s false-reject row is hatched throughout — the visual complement of the same table.

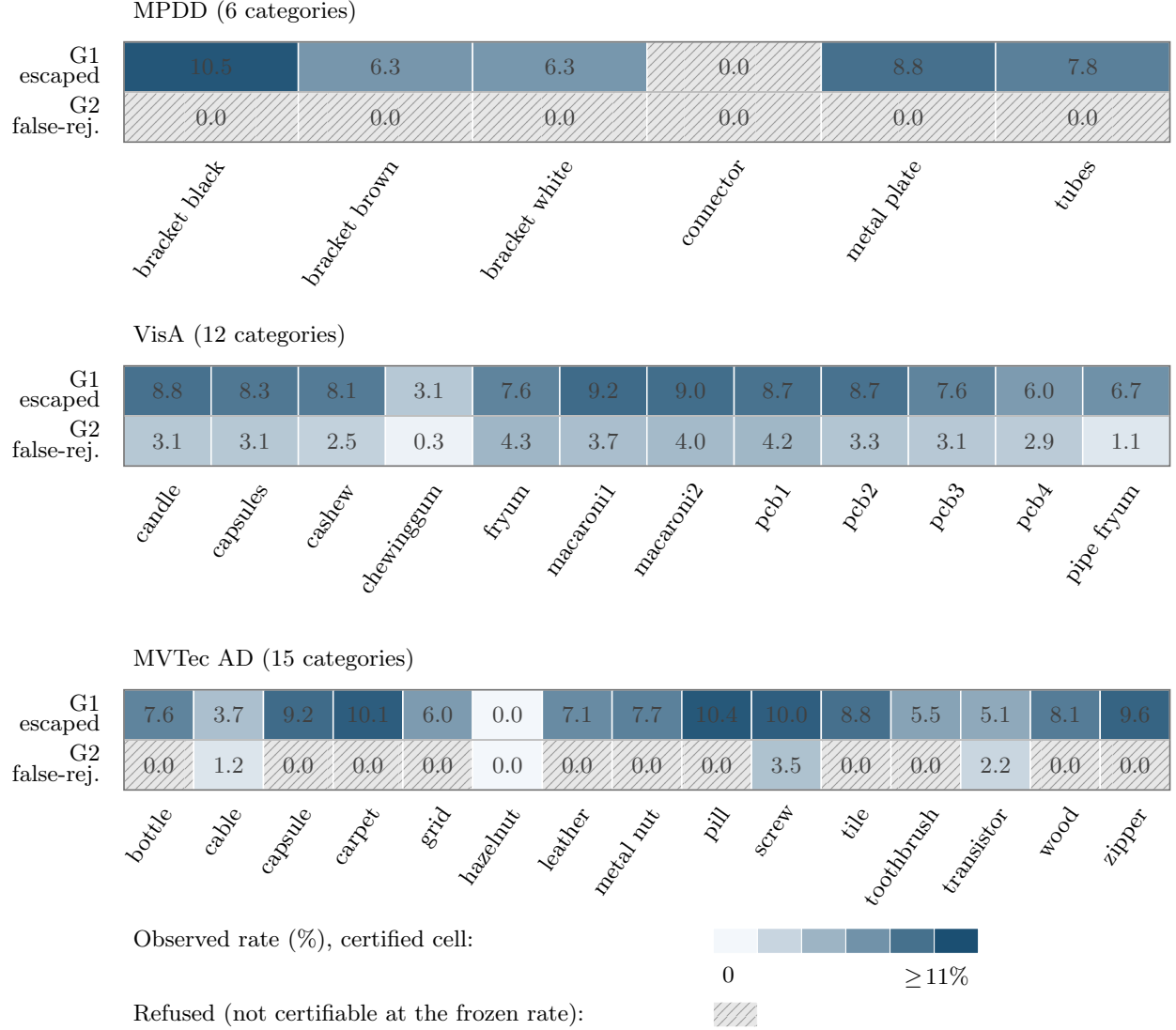


Figure 2: Per-category certification map for escaped-defect (G1) and false-reject (G2) at the frozen rates ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$), pooled over both backbones and five seeds, for MPDD, VisA, and MVTec AD. Colored cells are certified, shaded by the observed rate; hatched gray cells are refused as not certifiable at the frozen rate under the primary protocol (all of MPDD’s G2 column, MPDD’s connector on G1, and the 11/15 MVTec AD categories below the false-reject floor in Table 3).

5.4 Comparison with a conformal single-threshold baseline

The natural named baseline a reviewer would ask for is not a hand-picked straw man but the single-threshold construction our own G1 certificate specializes: split-conformal Conformal Risk Control (CRC) (Angelopoulos et al., 2024), applied with one per-category threshold at $\alpha_{\text{miss}} = 0.10$ and no defer option. For the 0/1 miss loss this threshold is exactly our G1 split-conformal threshold (an identity proved in the gate module’s implementation), so CRC and our gate share the same finite-sample escaped-defect guarantee and certify the same cells; the only degree of freedom is

what each does with the ambiguous middle. We recompute CRC on the identical canonical score dumps, under the identical protocol ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$, $R = 20$ repeated stratified 50/50 calibration/evaluation splits, five seeds, both backbones), so every cell in Table 4 is apples-to-apples with the main results. CRC has no mechanism to certify the false-reject side at all — with one threshold, every non-passed image is an auto-reject, not a deferral — so its G2 column is structurally n/a.

Table 4: Our dual gate vs. a single-threshold CRC baseline controlling the same escaped-defect risk ($\alpha_{\text{miss}} = 0.10$), pooled over both backbones, five seeds, and all categories ($n_{\text{cells}} = 2 \times 5 \times n_{\text{categories}}$ per benchmark). CRC certifies the same escaped-defect cells (G1) but has no false-reject certificate (G2 n/a) and must auto-reject the entire ambiguous band instead of deferring it.

Benchmark	Method	G1 cert.	G2 cert.	Escaped	False-reject	Deferral
MPDD	Our dual gate	50/60	0/60	6.6%	0.0%	73.1%
MPDD	CRC (single thr.)	50/60	n/a	6.6%	36.9%	0.0%
VisA	Our dual gate	120/120	120/120	7.6%	3.0%	16.4%
VisA	CRC (single thr.)	120/120	n/a	9.5%	16.2%	0.0%
MVTec AD	Our dual gate	150/150	40/150	7.3%	0.5%	54.4%
MVTec AD	CRC (single thr.)	150/150	n/a	8.5%	3.1%	0.0%

On MPDD, where the false-reject floor refuses G2 entirely (Section 5.3), the contrast is starkest: CRC pays 36.9% false-reject to hold the same 6.6% escaped-defect rate our gate holds at 0.0% false-reject, by deferring 73.1% of images instead of auto-rejecting them. VisA and MVTec AD show the same direction at smaller magnitude (16.2% vs. 3.0% false-reject on VisA; 3.1% vs. 0.5% on MVTec AD) — CRC’s realized escaped-defect rate is also somewhat higher on these two benchmarks (9.5% vs. 7.6% on VisA, 8.5% vs. 7.3% on MVTec AD), consistent with both methods controlling the same risk to within realized-rate sampling noise across repeats rather than producing bit-identical per-repeat routes. The comparison isolates exactly what the dual-gate design buys over the nearest published single-threshold alternative: not a better escaped-defect guarantee (both share the same conformal construction) but a false-reject guarantee CRC cannot offer, purchased by turning auto-rejects into deferrals. Figure 3 plots the three rates for both methods side by side across all three benchmarks, making the MPDD contrast in particular immediately visible.

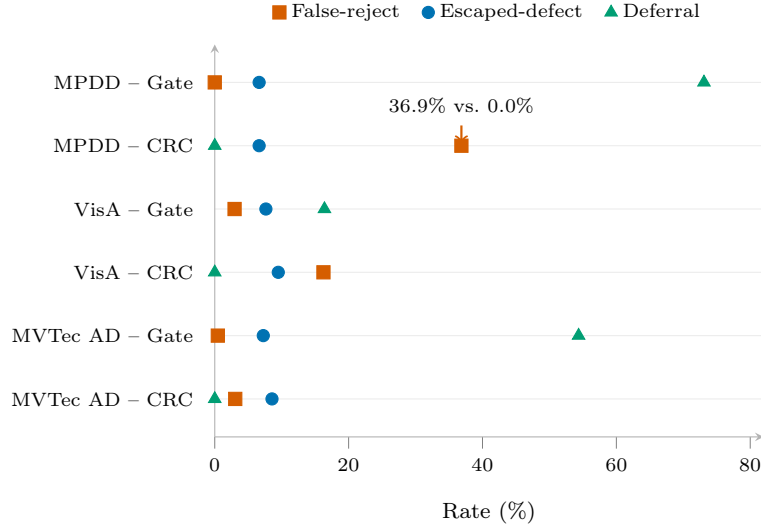


Figure 3: Escaped-defect, false-reject, and deferral rates for our dual gate versus the single-threshold CRC baseline (Table 4), pooled over both backbones and five seeds. CRC has no deferral mechanism, so every non-passed image becomes an auto-reject; on MPDD this drives its false-reject rate to 36.9% against 0.0% for our gate at the same escaped-defect rate.

As a second, non-conformal reference line we also compute the field-standard selective-prediction risk-coverage curve (Geifman and El-Yaniv, 2017) (per-category best-F1 threshold, margin-based abstention) on the same score dumps. Its area-under-the-risk-coverage-curve (AURC, lower is better) ranges from 0.0019 (MVTec AD, Dinomaly) to 0.0849 (VisA, PatchCore); at $\sim 80\%$ coverage the realized risk is 0.0625/0.0154 (MPDD, PatchCore/Dinomaly), 0.1315/0.0180 (VisA, PatchCore/Dinomaly), and 0.0095/0.0026 (MVTec AD, PatchCore/Dinomaly). This is a fundamentally different guarantee — a curve traced out by sweeping coverage post hoc, not a fixed pre-specified certificate at a chosen risk level — so we report it as context rather than a like-for-like comparison to Table 4.

5.5 Deployment cost analysis

Table 4 certifies the dual gate’s rates but not their economic consequence: every deferral is a decision routed to a human reviewer, and every false-reject the gate avoids is one the CRC baseline pays for instead. We translate both operating points into practitioner economics under three ASSUMED, explicitly labeled illustrative deployment scenarios spanning a $30\times$ throughput range — a low-volume precision line (100 parts/hour, \$5,000 per escaped defect, \$150 per false reject, a \$45/hour reviewer at 120 parts/hour), a mid-volume general industrial line (600/hour, \$300/\$15, a \$30/hour reviewer at 240/hour), and a high-volume commodity line (3,000/hour, \$25/\$2, a \$26/hour reviewer at 450/hour) — applied to the certified rates of Table 4 (neither the rates nor the frozen calibration are altered; only the dollar figures are scenario-dependent). The full nine-cell benchmark $\times$ scenario grid and script are in the released artifact set of the code-and-data deposit; Table 5 shows the mid-volume scenario.

Table 5: Deployment cost analysis, mid-volume general industrial scenario (600 parts/hour, \$300 per escaped defect, \$15 per false reject, \$0.125/part review labor). Reviewer load is throughput $\times$ deferral rate; CRC’s is 0 by construction (no deferral mechanism — every non-passed item is an automatic reject, not a human decision). Breakeven review cost/part is the human-review unit cost above which CRC would become cheaper than the dual gate, holding this scenario’s other costs fixed; headroom is that breakeven divided by the \$0.125/part assumed here.

Benchmark	Reviewer load	Gate \$/hr	CRC \$/hr	Gate savings	Headroom
MPDD	438.5/hr (1.83 FTE)	\$11,990	\$15,252	+\$3,262/hr	60 $\times$
VisA	98.4/hr (0.41 FTE)	\$14,024	\$18,547	+\$4,522/hr	369 $\times$
MVTec AD	326.1/hr (1.36 FTE)	\$13,134	\$15,661	+\$2,527/hr	63 $\times$

The dual gate is the cheaper deployment in all nine benchmark $\times$ scenario combinations we compute. At the low-volume precision line, MPDD’s 73.1% deferral becomes 73.1 parts/hour to a single part-time reviewer, saving \$5,500/hour against CRC’s 36.9% false-reject rate. At the high-volume commodity line the same deferral becomes 2,192.7 parts/hour (nearly five reviewer FTEs), yet the gate still saves \$2,084/hour net. The saving is not an artifact of favorable escaped-defect rates: on MPDD the two methods share the identical escaped-defect rate by the proven $G1 \equiv CRC$ threshold identity, and on VisA/MVTec AD the gate’s escaped rate is additionally somewhat *lower* than CRC’s. The least robust cell is MVTec AD under the high-volume commodity scenario: human review would need to cost more than \$0.69/part before CRC becomes cheaper there, about 12 $\times$ the \$0.058/part assumed in that scenario. Every other cell has 17–1,833 $\times$ headroom. This model omits reviewer error rate, queueing dynamics, and an appeal channel for CRC’s false-rejected good parts; modeling the latter would raise CRC’s cost further, so the comparison is if anything conservative toward CRC. Its dollar figures are scenario-dependent by construction, while the throughput-invariant breakeven quantities above are the assumption-robust takeaway.

5.6 G2 train-holdout promotion on MVTec AD (frozen remedy arm)

The preregistered calibration-efficiency remedy — recalibrating G2 on held-out *train-good* scores, gated behind a per-category KS exchangeability test with a mandatory audited-not-certified fallback — was computed after the freeze, for PatchCore only: Dinomaly has no train-side score dump (its reference eval scores test images only), so the promotion arm is single-backbone by construction, disclosed here rather than implied otherwise. It behaves as preregistered: G2-certifiable categories rise from 4/15 to 13/15 in seeds 0–3 and 12/15 in seed 4. The KS gate does real work rather than rubber-stamping: leather fails it in all five seeds (its train-good and test-good score distributions are not exchangeable) and screw additionally fails in seed 4 — each is reported audited-not-certified, never silently promoted, and the seed-4 screw case shows a category certified under the primary protocol can still be *refused* under the holdout arm when exchangeability breaks. Toothbrush remains uncertifiable at $\alpha_{\text{fr}} = 0.05$ under both arms, exactly as preregistered (n^{good} too small in either pool). K1 is not tripped in any promoted seed aggregate.

5.7 MPDD train-holdout rescue (post-freeze exploratory)

We report the primary-protocol 0/6 of the preceding section as MPDD’s thesis-confirming headline; the train-holdout arm below is a clearly secondary, post-freeze, PatchCore-only result. The train-holdout calibration-efficiency lever the frozen MVTec remedy exercises (Section 5.6) turns MPDD’s

0/6 primary refusal into a 4/6 rescue, a vivid single-benchmark illustration of the same lever. MPDD’s train-good pools are large (54–289 per category) where its test-good pools are tiny, so recalibrating G2 on a 20% held-out train-good pool (the flag-gated train-holdout calibration mode, 20% holdout fraction; PatchCore only, as on MVTec — Dinomaly dumps no train-side scores, so the arm is single-backbone by construction) clears the false-reject floor for the 5/6 categories whose holdout pool reaches $n \geq 19$. The same per-category KS exchangeability gate then adjudicates each promotion, and it does decisive work: G2-certifiable rises from 0/6 to 4/6, identical in all five seeds. Two categories are honestly held back — metal plate, whose holdout pool (11) is still below the floor, and tubes, which fails the KS gate (BH-adjusted $p \approx 1.2 \times 10^{-5}$; its train-good and test-good score distributions are demonstrably non-exchangeable) and is reported audited-not-certified rather than promoted. The remaining four (bracket black, bracket brown, bracket white, connector) promote cleanly (KS $p_{\text{BH}} \geq 0.44$ every seed). Connector is the one category G1 refuses under the primary protocol ($n_{\text{cal}}^{\text{def}} = 7 < 9$ defective) yet the arm G2-certifies (holdout-good $26 \geq 19$): the two certificates gate on independent sample axes — defect count for escaped-defect, good count for false-reject — so this is coherent, not contradictory. Its 100% deferral (the highest of any category, since G1 refusal leaves the auto-reject band empty) nonetheless makes that G2 certificate operationally hollow in practice. So the count-only floor of 5/6 is audited down to a genuinely certified 4/6 — the KS guardrail keeping the rescue honest, exactly the exchangeability risk MPDD’s non-homogeneous backgrounds were expected to pose. As a sanity check (no gate), the holdout arm’s 80%-memory-bank test scores rank-correlate with the primary 100%-bank scores at Spearman $\rho \geq 0.918$ across all 6×5 category-seed cells (max $|\Delta \text{image-AUROC}| = 0.040$), confirming the same model was scored. This arm is post-freeze exploratory and flag-gated prereg-neutral (it re-freezes nothing and is never in the confirmatory Holm family); the 0/6 $\rightarrow$ 4/6 contrast on the same benchmark and the same scores is a vivid single-benchmark demonstration of the calibration-efficiency lever. The larger, *confirmatory* instance of the same lever is the frozen MVTec train-holdout remedy (Section 5.6), which lifts G2 from 4/15 to 12–13/15 across seeds (13 in seeds 0–3, 12 in seed 4). The primary-protocol 0/6 stays the honest headline for this industrial benchmark: on real metal parts the gate certifies nothing at the 5% false-reject budget because the data cannot pay for it — exactly what a refusal-first design should do.

5.8 Validity audit V1 (tier-1) and the kill-gates

Per frozen amendment A3, tier-1 is reported *per axis*, with the false-reject axis evaluated only over G2-certified cells: wherever G2 is refused the auto-reject region is empty and the measured false-reject rate is identically 0 — a pass without evidence that A3 excludes from every count. On the escaped-defect axis, tier-1 passes in all 150 MVTec cells (2 backbones $\times$ 5 seeds $\times$ 15 categories), all 120 VisA cells ($2 \times 5 \times 12$), and all 60 MPDD cells ($2 \times 5 \times 6$), every rate within target+3pp. On the false-reject axis, MVTec has 40 evaluable cells (4 G2-certified categories $\times$ 2 $\times$ 5) and passes 40/40, with the other 110 cells reported G2-REFUSED and excluded; on VisA every category is G2-certified, so all 120/120 cells are evaluable and pass non-vacuously; on MPDD every category is G2-refused under the primary protocol, so all 60 false-reject cells are G2-REFUSED and excluded (the mirror image of VisA, and the primary-protocol counterpart to the train-holdout rescue of Section 5.7). Tier-1 is a necessary-not-sufficient pipeline check — a conservative gate realizes rates well below target, so passing validates the calibrate/route/evaluate machinery rather than demonstrating sharpness. Consequently K1 is not tripped (the A3 exclusion cannot change K1’s count: excluded cells cannot fail) (0 violations in every (backbone, seed) aggregate, against the 5-cell

MVTec/VisA kill threshold and, on MPDD, against a proportionally-scaled 2-cell threshold²) and K2 is not tripped (on MVTec/VisA at most 1 vacuous category — VisA/PatchCore/seed 2 — against the 8-category/0.80-deferral threshold; on MPDD at most 2 vacuous categories per aggregate against the scaled 4-category threshold), in all 30 aggregates across the three benchmarks (Table 7). Table 6 reports seed-0 median per-category deferral, its mean where computed, and the extreme categories, for each benchmark and backbone. MVTec AD’s deferral is elevated because G2 refusal empties the auto-reject band in 11/15 categories, though still far from the vacuity threshold. VisA’s deferral drops sharply because G2 certifies every category there, so the auto-reject band is live throughout — direct operational evidence that the elevated MVTec deferral is the price of honest G2 refusal, not a property of the gate. MPDD sits at the far end: G2 is refused in all 6 categories under the primary protocol, exactly what 0/6 certification predicts. These are three honest operating points, not a defect and a fix: the elevated MVTec and MPDD deferral is the cost of refusing to certify small good-calibration pools, and VisA’s low deferral is what the same gate, same code path, spends when the good pools are large enough to certify every category.

Table 6: Seed-0 median per-category deferral by benchmark and backbone (Section 5.8), with the mean where computed and the extreme categories. G2 refusal (Section 5.3) empties the auto-reject band and elevates deferral on MVTec AD and MPDD; VisA’s universal G2 certification keeps deferral low.

Benchmark	Backbone	Median	Mean	Max category	Min category
MVTec AD	PatchCore	70.0%	55.1%	pill 78.0%	transistor 2.0%
MVTec AD	Dinomaly	69.5%	54.2%	capsule 77.3%	screw 3.1%
VisA	PatchCore	18.7%	—	macaroni2 72.5%	chewinggum 2.0%
VisA	Dinomaly	4.2%	—	capsules 21.9%	candle 2.5%
MPDD	PatchCore	74.1%	—	connector 100%	bracket brown 61.8%
MPDD	Dinomaly	70.9%	—	connector 100%	bracket white 53.3%

²At $n_{\text{cat}} = 6$ the design’s absolute kill thresholds (max 5 violations, min 8 categories) are vacuously un-trippable, so on MPDD they are scaled to the design’s proportions — $5/15 \rightarrow \lceil 6/3 \rceil = 2$ for K1 and $8/15 \rightarrow \lceil 6 \cdot 8/15 \rceil = 4$ for K2 — with the raw counts recorded so any threshold is re-derivable. The no-trip verdict is invariant to the rounding rule: any reasonable rounding of the proportional thresholds gives the same outcome (0 violations and ≤ 2 vacuous categories clear even the tightest rounding). These diagnostic kill-gates are never part of the confirmatory family.

Table 7: V1 tier-1 (per axis, per frozen amendment A3) and kill-gates, aggregated over the 30 (backbone, seed) cells across all three benchmarks. [†]False-reject evaluated only over the 4 G2-certified MVTec categories; the other 110 cells are G2-REFUSED and excluded from the count, never counted as (vacuous) passes. [‡]On MPDD all 60 false-reject cells are G2-REFUSED under the primary protocol and excluded (see Section 5.7 for the train-holdout rescue). MPDD’s K1/K2 thresholds are proportionally scaled to $n_{\text{cat}} = 6$ (2 and 4). “—” marks a cell that does not apply to that row (the raw evaluable-cell counts above have no separate threshold or trip verdict; K5 never fires under the frozen design, so it reports no single value). Sources: the frozen MVTec AD, VisA, and MPDD analysis summaries and their per-cell validity dumps; frozen preregistration §9 amendment A3.

Axis	Value	Threshold	Tripped?
V1 tier-1 escaped-defect cells (MVTec AD)	150/150	—	—
V1 tier-1 false-reject cells (MVTec AD) [†]	40/40	—	—
V1 tier-1 escaped-defect cells (VisA)	120/120	—	—
V1 tier-1 false-reject cells (VisA, all G2-certified)	120/120	—	—
V1 tier-1 escaped-defect cells (MPDD)	60/60	—	—
V1 tier-1 false-reject cells (MPDD) [‡]	0 evaluable	—	—
K1 coverage violations (per cell)	0	≥ 5 (≥ 2 MPDD)	No (all 30)
K2 vacuous categories (per cell)	0 (1 in VisA/ PatchCore/seed 2; ≤ 2 on MPDD)	≥ 8 (≥ 4 MPDD)	No (all 30)
K3 backbone floor (mean AUROC)	≥ 0.982 (MVTec) / ≥ 0.903 (VisA) / ≥ 0.948 (MPDD)	< 0.90	No
K5 calibration floor	—	$\alpha_{\min} > 0.10$ (≥ 3 cat)	Cannot fire

5.9 Exploratory excess-AURC audit

As an exploratory (not confirmatory) readout scoped to one (backbone, seed, category) family of size 2 (practices B1 and B2; B3 could not run — see below), on the repeat-0 split at seed 0: on MVTec AD, PatchCore rejects the random-deferral null in 10/30 practice-cells after within-cell Holm, with 6/30 degenerate (per-category AUROC = 1.0 $\Rightarrow$ zero measurable headroom); Dinomaly rejects in 0/30 with 14/30 degenerate — an expected ceiling effect, since Dinomaly sits at per-category AUROC 1.0 in 8/15 categories on this split, leaving threshold practices no room to show skill above a near-perfect baseline. VisA resolves this ceiling effect: on the same seed-0 readout, PatchCore rejects in 14/24 cells and Dinomaly in 16/24, with 0 degenerate cells for either backbone — on a benchmark where the backbones are not saturated, field-standard threshold practices demonstrably do earn deferral skill over honest random deferral, so the audit’s verdict axis is live rather than vacuously constructive. Across all five seeds the picture is stable (MVTec: PatchCore 48/150, Dinomaly 4/150; VisA: PatchCore 64/120, Dinomaly 42/120 practice-cell rejections). One comparability caveat: on VisA, Dinomaly is a multi-class model (one model per seed for all 12 categories) while PatchCore keeps per-category memory banks, so cross-backbone audit comparisons there are not architecture-controlled and are read per backbone, not head to head.

5.10 Confirmatory audit verdict (C2)

The exploratory readout above pools nothing; its higher-powered confirmatory counterpart pools the 15 MVTec categories per (practice, backbone). The confirmatory family is the four tests $\{\text{B1 fixed, B2 tuned}\} \times \{\text{PatchCore, Dinomaly}\}$; the train-good quantile practice B3 drops out because the frozen confirmatory pass loaded no train-good scores, the degradation from six to four tests preregistered in PREREG §4. B3 is not infeasible in general: a held-out train-good score pool does exist for PatchCore, so its B3 arm is completed post-hoc below; Dinomaly, however, has no train-side score dump at all, so the full six-member family cannot be run even now. For each (practice, backbone) we pool the 15 categories into one evaluation half (repeat-0 split), fit B1 on the pooled calibration half and B2 per category, and test the excess area under the risk-coverage curve against the closed-form random-deferral null with a matched-abstention permutation test ($n_{\text{perm}} = 2000$, strata = category) and a category-blocked bootstrap CI. Every one of the four tests rejects the random-deferral null at the permutation floor $p = 1/2001 = 5 \times 10^{-4}$ (Holm-adjusted $p = 2 \times 10^{-3}$) in all five backbone seeds, with pooled excess-AURC in $[0.023, 0.050]$ and every bootstrap CI excluding zero (Table 8). The frozen confirmatory construction is the per-seed four-member Holm family, and it rejects in every one of the five seeds. The preregistration is silent on how to combine seeds, so we report the seed dimension as a robustness check rather than a family axis: the single cross-seed rollout (the seed-max of the per-seed p -values) was authored post-freeze as a one-shot convenience and is moot here, the verdict being identical under seed-min, seed-median, and seed-max. The constructive arm of C2 publishes: field-standard threshold practice earns real deferral skill over honest random deferral. Pooling recovers across the 15 categories the power that per-category saturation denies the exploratory single-category readout.

B3 (train-good quantile), the family’s feasible half, post-hoc. Because a held-out train-good score pool exists for PatchCore (the holdout run scored the per-category train-good images alongside the test set), we complete the third practice for that backbone post-hoc, on the same pooled-category construction. B3 is constructive and consistent with B1/B2: across all five seeds it rejects the random-deferral null at the permutation floor $p = 5 \times 10^{-4}$, with pooled excess-AURC in $[0.053, 0.056]$ (larger than B1’s and B2’s) and every category-blocked bootstrap CI excluding zero; a cross-run sensitivity arm agrees. Added to the frozen four-member per-seed family as a fifth member, all five reject in all five seeds (Holm $p = 2.5 \times 10^{-3}$). This is labelled post-hoc and exploratory: the confirmatory verdict remains the frozen four-member family above, and B3-Dinomaly stays impossible for want of a train-side dump.

On VisA (post-freeze, exploratory, and kept out of the MVTec confirmatory Holm family) the same pooled construction rejects the random-deferral null in all four practice $\times$ backbone tests across all five seeds (Holm $p = 2 \times 10^{-3}$), with larger pooled excess-AURC $[0.044, 0.106]$ — the headroom the near-saturated MVTec backbones deny (Table 8, lower block).

Table 8: Pooled excess-AURC audit (C2). Family = {B1 fixed, B2 tuned} $\times$ {PatchCore, Dinomaly}, Holm $\alpha = 0.05$. MVTec AD is the frozen confirmatory family: all four tests reject the random-deferral null in every one of the five backbone seeds, and the per-seed values are the frozen confirmatory verdict. VisA (post-freeze) is reported exploratory and is kept out of the MVTec Holm family. Excess-AURC is the seed range; permutation floor $p = 1/2001$.

Benchmark	Backbone	Practice	excess-AURC (seed range)	perm. p	Holm p / reject
MVTec AD	PatchCore	B1 fixed	0.023–0.030	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	PatchCore	B2 tuned	0.036–0.046	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	Dinomaly	B1 fixed	0.045–0.050	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	Dinomaly	B2 tuned	0.027–0.035	5×10^{-4}	2×10^{-3} / ✓
VisA	PatchCore	B1 fixed	0.073–0.089	5×10^{-4}	2×10^{-3} / ✓
VisA	PatchCore	B2 tuned	0.086–0.106	5×10^{-4}	2×10^{-3} / ✓
VisA	Dinomaly	B1 fixed	0.081–0.095	5×10^{-4}	2×10^{-3} / ✓
VisA	Dinomaly	B2 tuned	0.044–0.055	5×10^{-4}	2×10^{-3} / ✓

Verdict: constructive arm on both benchmarks. All four MVTec tests reject in all five seeds (frozen confirmatory); all four VisA tests reject in all five seeds (post-freeze exploratory). Every bootstrap CI excludes 0.

5.11 When a fixed threshold over-promises (post-hoc binding demonstration)

The excess-AURC audit shows practice earns deferral skill, but it does not directly exhibit the sharper event: a category where a *naive fixed threshold* actually violates a target error rate while the certified gate holds. We test that event directly, post-hoc. For each (backbone, category) we take the naive practitioner’s operating point — a single global best-F1 threshold (B1) applied to every eval image with no deferral — and compute its realized escaped-defect and false-reject rates on the held-out eval half; alongside it we compute the certified gate’s realized rates and its deferral cost on the same split (mean over the $R = 20$ repeats, at backbone seed 0). A cell *binds* when the fixed threshold’s realized rate exceeds the target ($\alpha_{\text{miss}} = 0.10$ or $\alpha_{\text{fr}} = 0.05$) while the gate’s stays within target+3pp.

Such cells exist, on both benchmarks, both axes, and both backbones. Restricting to the strongest evidence — cells where the gate is certified on that axis in all 20 repeats and the bind holds in all five backbone seeds — there are 27: on MVTec AD, 3 escaped-axis (Dinomaly on capsule, pill, screw) and 3 false-reject-axis (PatchCore on screw; Dinomaly on cable, transistor); on VisA, 4 escaped-axis and 17 false-reject-axis cells. Crucially the effect is not an artifact of the gate abstaining on everything: the cleanest cases pay a small deferral cost. On MVTec AD a global threshold lets 24.2% of defective screws escape; the certified gate, deferring just 4% of images, holds the escaped rate at 8.0% (under target) because its per-category (Mondrian) calibration puts the auto-pass boundary where a global threshold cannot. On the false-reject axis, the same global threshold auto-rejects 19.5% of good screws while the gate holds false-reject at 5.0% (certified, 14% deferral); on VisA the extreme case is capsules, where a global threshold would auto-reject 96.8% of good parts and the gate holds false-reject at 3.7%. The mechanism the demonstration exposes is that a single fixed *global* threshold cannot serve categories with different score scales at once, so it over-promises on whichever it mis-fits; the certified gate refuses (defers) exactly where it cannot honor the target and calibrates per category elsewhere. This is a post-hoc, exploratory illustration — not a preregistered arm — and is reported verdict-honestly: had no cell bound, we would have kept the “not demonstrated” framing.

5.12 V1 tier-2 verdict

Under the frozen amendments the tier-2 power floor is applied per repeat, not pooled over repeats (amendment A1). On MVTec AD the escaped-defect axis is per-repeat powered in 13/15 categories (underpowered in toothbrush and transistor, per-repeat $n_{\text{eval,def}} = 15, 20 < 22$). Graded on a per-repeat one-sided 95% Clopper–Pearson upper bound against the 0.13 threshold, the powered cells overwhelmingly fail (115 fail / 15 pass out of 130 powered, 20 excluded as underpowered: toothbrush, transistor; the 15 passes are the near-zero-miss cells — Dinomaly on cable and hazelnut, PatchCore on hazelnut). This is the preregistered expectation of amendment A2, not a certificate failure: split-conformal calibration targets α_{miss} exactly, so a one-sided upper bound at a per-repeat $n \approx 30\text{--}70$ sits above the target-plus-3pp tolerance whenever the realized miss rate is near the target. The certificate-validity statistic is tier-1, which passes in all 150 MVTec cells; tier-2 is a stringent estimator-variance readout layered on top (Table 9 gives the per-category breakdown). The tier-2 false-reject axis is *structurally ungraded* on MVTec AD: it is scored only over G2-certified categories (amendment A3), and those four (cable, hazelnut, screw, transistor) are themselves per-repeat underpowered ($n_{\text{eval,good}} \leq 30 < 36$), so 0 pass / 0 fail out of 150 cells are gradeable (110 G2-refused + 40 underpowered excluded) — exactly the amendment A1+A3 outcome. One caveat we disclose: the power partition and the false-reject buckets are exact (deterministic from the cached counts), but the escaped pass/fail split within powered cells is reconstructed from a per-repeat Clopper–Pearson upper bound built on the pooled rate as the per-repeat point estimate, not from the raw per-repeat records. On VisA (post-freeze, exploratory) the larger good pools clear the false-reject power floor in part, so that axis is partially gradeable there: escaped 17 pass / 103 fail (of 120), false-reject 4 pass / 66 fail / 50 underpowered (of 120).

Table 9: V1 tier-2 grading, per category (expanded from the aggregate axis totals in the text; every cell out of 10 = 2 backbones $\times$ 5 seeds). Escaped-defect is pass/fail/unpowered against the per-repeat 0.13 Clopper–Pearson bound (amendment A2); false-reject is pass/fail/unpowered/refused, where “refused” means the category is not G2-certified (amendment A3) and “unpowered” means it is G2-certified but fails the per-repeat power floor (amendment A1). MVTec AD is 0-gradeable on false-reject by construction (no G2-certified category clears the power floor); VisA (post-freeze, exploratory) is partially gradeable.

Category	Escaped-defect (pass/fail/unpwr.)	False-reject (pass/fail/unpwr./refused)
<i>MVTec AD (confirmatory tier-1 family; amendments A1/A2/A3)</i>		
bottle	0/10/0	0/0/0/10
cable	5/5/0	0/0/10/0
capsule	0/10/0	0/0/0/10
carpet	0/10/0	0/0/0/10
grid	0/10/0	0/0/0/10
hazelnut	10/0/0	0/0/10/0
leather	0/10/0	0/0/0/10
metal_nut	0/10/0	0/0/0/10
pill	0/10/0	0/0/0/10
screw	0/10/0	0/0/10/0
tile	0/10/0	0/0/0/10
toothbrush	0/0/10	0/0/0/10
transistor	0/0/10	0/0/10/0
wood	0/10/0	0/0/0/10
zipper	0/10/0	0/0/0/10
<i>VisA (post-freeze, exploratory)</i>		
candle	0/10/0	0/10/0/0
capsules	0/10/0	0/0/10/0
cashew	0/10/0	0/0/10/0
chewinggum	10/0/0	0/0/10/0
fryum	0/10/0	0/0/10/0
macaroni1	0/10/0	0/10/0/0
macaroni2	0/10/0	0/10/0/0
pcb1	0/10/0	0/10/0/0
pcb2	0/10/0	0/10/0/0
pcb3	0/10/0	0/10/0/0
pcb4	5/5/0	4/6/0/0
pipe_fryum	2/8/0	0/0/10/0
Totals		
MVTec AD (150 cells)	15/115/20	0/0/40/110
VisA (120 cells)	17/103/0	4/66/50/0

5.13 Latency and deployment cost

The gate is a thin decision layer on top of whatever detector already produces the anomaly score, so the only cost it adds is the conformal calibration (once, per calibration refresh) and the per-image routing comparison. We measured both on this work’s hardware, and we place them alongside the backbone forward pass that dominates the end-to-end budget (Table 10). Routing an image is an $O(1)$ threshold comparison inside an $O(\text{sort})$ calibration; the measured per-image routing cost is $2.8\,\mu\text{s}$ on CPU — roughly four orders of magnitude below the backbone forward pass and under 0.02%

Table 10: Per-image latency and deployment cost. The two gate rows and the Dinomaly backbone row are *measured on this work’s hardware* (gate: Intel Core Ultra 9 275HX CPU, 24 threads; Dinomaly: NVIDIA RTX 5090 Laptop GPU, torch 2.12.0, batch 1, median over 250 real VisA test images, full forward + anomaly-map + image-score pass). The PatchCore backbone row is *reported by the authors* (not measured here; Roth et al., 2022, $\sim 1\%$ -coreset configuration on their hardware) and is included only to bracket the backbone budget — backbone timing is hardware- and configuration-dependent, whereas the gate overhead below it is the measured constant this paper contributes. Sources: the frozen latency measurement dumps.

Component	Per-image cost	Notes
<i>Backbone forward pass (dominates the budget)</i>		
Dinomaly (DINOv2-b, 148M) — measured, GPU	18.9 ms (p95 23.1)	53 img/s at batch 1; 17.0 ms/img batched ($B=32$)
PatchCore (WRN50, coreset) — <i>authors’ figure</i>	$\lesssim 200$ ms	Roth et al. (2022), their hardware; not measured here
<i>Gate overhead added on top (this paper) — measured, CPU</i>		
Routing (per image)	$2.8 \mu\text{s}$	$O(1)$ compare; $< 0.02\%$ of the Dinomaly forward pass
Calibration (one-time)	2.4 ms (p95 2.8)	full 3-way gate, 861-img cal half, 15 strata; per refresh

of Dinomaly’s per-image GPU inference. Calibrating the full three-way gate over one 861-image calibration half across all 15 MVTec strata (the $\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$ primary configuration) takes 2.4 ms end to end, and is amortized over an entire evaluation run rather than paid per image. The practical conclusion is that certification is free relative to detection: a deployment already able to afford the detector can add the escaped-defect / false-reject certificate at no measurable latency cost. The backbone rows are the honest budget the gate sits on — Dinomaly measured here on an RTX 5090 Laptop GPU, PatchCore’s row taken from the authors’ reported figure (not measured here, and both hardware- and coreset-dependent). We flag the substrate mix these numbers span, so no cross-row comparison is over-read: the gate rows are timed on CPU over the MVTec calibration scores (861-image cal half, 15 strata), the Dinomaly backbone row on GPU over 250 real VisA test images (the VisA branch-A seed-0 checkpoint), and the PatchCore row is the authors’ published $\sim 1\%$ -coreset figure whereas our scoring uses the 10%-coreset configuration — so the table brackets the order of magnitude of the backbone budget the gate rides on, and it is the μs -scale CPU gate overhead, not any CPU-vs-GPU or MVTec-vs-VisA backbone comparison, that the paper contributes.

6 Limitations and gated arms

The preregistration was frozen on 2026-07-11 (a sha256-pinned sign-off record over the preregistration document and its A1–A3 amendments). One frozen arm has since been computed and is reported above (the G2 train-holdout promotion, Section 5.6). The VisA and MPDD benchmarks are both post-freeze additions — the frozen preregistration mentions neither — run under the identical frozen protocol and code path but reported as exploratory, never confirmatory, throughout Section 5; MPDD’s train-holdout rescue (Section 5.7) is additionally a flag-gated, prereg-neutral protocol variant. The arms below were unlocked but not yet computed at freeze time. Each bullet states what has since been computed and cross-references its full report in Section 5 rather than repeating it.

- **Confirmatory audit family (C2).** Computed and reported in full in Section 5.10 (Table 8): constructive on MVTec AD in all five seeds, robust to the choice of seed-rollup.
- **B3 (train-good quantile) practice.** Completed post-hoc for PatchCore (Section 5.10), constructive and consistent with B1/B2. It remains impossible for Dinomaly (no train-side score dump on any benchmark) and unavailable on VisA for either backbone (no train-good scores were dumped there).
- **V1 tier-2 as a verdict.** Graded in full in Section 5.12 (Table 9): the MVTec escaped axis mostly fails under amendments A1/A2 — the expected estimator-variance behaviour, not a certificate failure — while the false-reject axis is structurally ungraded under A1+A3. VisA is partially gradeable on both axes.
- K6 (scoop re-scan). Now executed: the pre-submission citation re-scan for any prior certified three-way triage on MVTec AD or VisA ran fresh after the freeze and found no scoop (Section 2), so the pre-decided pivot to C2+C3-primary was not triggered. K6 is a *novelty* gate, not a statistical one, so running it last, after the freeze, is the intended sequencing, not an inconsistency.
- K4 (audit headroom). **[TODO/PREREG-GATED: K4 needs an oracle-headroom statistic the certification module does not yet implement.]**
- **Remaining tables/figures.** Table 8 and Table 9 are now included. **[TODO/PREREG-GATED: The defect-type Mondrian coverage-debt map (MVTec only; VisA has no defect-type axis) and the train-holdout G2 delta figure remain to be produced.]**

The latency/practicality slot flagged in the frozen skeleton is now computed and reported in full in Section 5.13 (Table 10), confirming the gate’s overhead is negligible next to backbone inference.

Two non-gated caveats are also worth stating plainly.

Substrate maturity on MVTec AD, and what VisA settles. Both reproduced backbones sit near ceiling on MVTec AD, leaving the exploratory excess-AURC audit little headroom to detect practice skill there (Section 5.9). VisA was the pre-freeze skeleton’s designed answer to exactly this ceiling effect (a lower-ceiling second benchmark, stated there as a plan; VisA itself is not part of the frozen preregistration), and it behaved as designed: the same audit is fully informative on VisA (Section 5.9). The confirmatory pooled audit (Section 5.10) and the post-hoc binding demonstration (Section 5.11) settle the deferral-skill question on MVTec AD directly. We state plainly that all three benchmarks are academic: no production-line, temporal-drift, or real-factory data backs the industrial framing here (threshold-transfer across production periods is future work), so the paper’s claims are about certified triage on the standard inspection benchmarks, not about a deployed line.

Small good-calibration pools are intended behavior, not a shortfall. G2’s certifiable count is a direct property of each benchmark’s good-pool size, not a conservatism baked into the gate: MVTec AD (4/15), VisA (12/12, identical code path), and MPDD (0/6 under the primary protocol, rescued to 4/6 under the flag-gated train-holdout arm) form three points consistent with pool-size-driven refusal (Sections 5.3, 5.7). MPDD’s rescue is PatchCore-only (Dinomaly dumps no train-side scores) and never enters the confirmatory family.

The $\alpha_{\text{miss}} = 0.10$ tolerance and industrially-tight escape rates. The near-universal G1 certification reported above (15/15 MVTec AD, 12/12 VisA, 5/6 MPDD) holds at $\alpha_{\text{miss}} = 0.10$ — a 10% chance of auto-passing a defective part, well outside what a safety-critical line would accept — and is an artifact of that tolerance being easy to clear against the $\alpha_{\text{min}} = 1/(n_{\text{cal}}^{\text{def}} + 1)$ floor at these benchmarks’ defect-calibration pools ($n_{\text{cal}}^{\text{def}} \in [15, 70]$ on MVTec AD, Table 3; $= 50$ on VisA; $\in [7, 36]$ on MPDD). At a deployable tolerance of $\alpha_{\text{miss}} = 0.01$ the floor requires $n_{\text{cal}}^{\text{def}} \geq 99$, which no category on any of the three benchmarks reaches, so G1 refuses in all 33/33 (benchmark, category) cells; at $\alpha_{\text{miss}} = 0.001$ ($n_{\text{cal}}^{\text{def}} \geq 999$) the same is true. We state this plainly as a scope boundary rather than leave it unfronted: on academic benchmarks of this size, the meaningful demonstration is the refusal discipline itself — that the gate honestly refuses rather than assert an escape guarantee the data cannot support — not a certified industrially-tight escape rate. Certifying escaped-defect at the 1%–0.1% tolerances a safety-critical line would require needs the larger labeled-defective pools (hundreds to thousands per category) that accrue from rework and scrap logging over months of production (Section 3.1), not the few dozen defective images an academic test split provides.

7 Conclusion

We described a certified three-way triage gate that turns any anomaly detector’s scores into an auditable inspection decision with finite-sample escaped-defect and false-reject guarantees, and a refusal mechanism that makes “not enough data to certify” a visible outcome. On MVTec AD, VisA, and MPDD, across two backbones and five seeds each, the calibration machinery matches its preregistered arithmetic exactly, with no kill-gate tripped in any of the 30 (backbone, seed) aggregates. The three benchmarks exercise complementary failure modes and together give three points consistent with pool-size-driven refusal ($0/6 \rightarrow 4/15 \rightarrow 12/12$): MVTec AD stresses the refusal side (G2 certifiable in only 4/15 categories under the primary protocol, lifted to 12–13/15 by the frozen train-holdout remedy — PatchCore-only by data availability — with KS-failing categories honestly refused); VisA — lower-ceiling, larger good pools — certifies both axes in all 12 categories, cuts median deferral to 4–19%, and gives the excess-AURC audit the headroom MVTec’s saturated backbones deny it; and MPDD (real painted-metal parts, post-freeze exploratory) is the stingy extreme, refusing G2 in all 6 categories under the primary protocol and rescued to 4/6 by the flag-gated train-holdout arm (one category KS-audited out). The confirmatory pooled audit (C2) returns a constructive verdict on MVTec AD — field-standard threshold practice earns real deferral skill over honest random deferral in all five backbone seeds (Section 5.10).

Data and code availability

MVTec AD (Bergmann et al., 2019), VisA (Zou et al., 2022), and MPDD (Ježek et al., 2021) (staged from the credential-free HuggingFace mirror `meksamiao/mpdd`, archive sha256 `69f8da73...ddd6` recorded in the frozen split manifest) are public benchmarks. The frozen analysis pipeline that produced every reported number — separate MVTec AD, G2-promotion, VisA, and MPDD (primary + train-holdout) analysis stages, each reading only the local score dumps and emitting a frozen result summary — together with those score dumps, the frozen preregistration, and its sha256-pinned sign-off, will be archived with the code below. The gate is implemented in *inspect-gate*, a thin wrapper over the project’s reliability metrics library. Per Springer Nature’s code-sharing policy (a bare repository link does not by itself carry a permanent identifier), the source is maintained in the `tools/inspect-gate` directory of the private GitHub repository `PeterPonyu/reliability-commons`

(commit `ac8ec55`), to be made public on acceptance, as the development mirror, and archived at Zenodo (DOI: 10.5281/zenodo.21392291, reserved; deposit publishes at submission) for the frozen score dumps, analysis code, and preregistration archive.

References

- Samet Akcay, Dick Ameln, Ashwin Vaidya, Barath Lakshmanan, Nilesch Ahuja, and Utku Genc. Anomalib: A deep learning library for anomaly detection. *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2022.
- Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.
- Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *arXiv:2110.01052*, 2021.
- Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *Proc. Int. Conf. Learning Representations (ICLR)*, 2024. *arXiv:2208.02814*.
- Songran Bai, Xiaolong Zheng, and Daniel Dajun Zeng. CRC-SGAD: Conformal risk control for supervised graph anomaly detection. *arXiv:2504.02248*, 2025.
- Stephen Bates, Anastasios Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6), 2021.
- Kilian Batzner, Lars Heckler, and Rebecca König. EfficientAD: Accurate visual anomaly detection at millisecond-level latencies. In *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2024.
- Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. MVTec AD—a comprehensive real-world dataset for unsupervised anomaly detection. In *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019.
- Charles J. Clopper and Egon S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Jia Guo, Shuai Lu, Weihang Zhang, Fang Chen, Huiqi Li, and Hongen Liao. Dinomaly: The less is more philosophy in multi-class unsupervised anomaly detection. In *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2025. *arXiv:2405.14325*.
- Oliver Hennhöfer, Maximilian Kirsch, and Christine Preisach. Conformal anomaly detection in Python: Moving beyond heuristic thresholds with `nonconform`. *arXiv:2605.13642*, 2026.
- Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- Štěpán Ježek, Martin Jonák, Radim Burget, Pavel Dvořák, and Miloš Skoták. Deep learning-based defect detection of metal parts: evaluating current methods in complex conditions. In *Proc. 13th Int. Congr. Ultra Modern Telecommunications and Control Systems and Workshops (ICUMT)*, pages 66–71, 2021. MPDD (Metal Parts Defect Detection) benchmark.

- Divake Kumar, Nastaran Darabi, Sina Tayebati, and Amit Ranjan Trivedi. Beyond confidence: Adaptive abstention in dual-threshold conformal prediction for autonomous system perception. In *Proc. IEEE Int. Conf. Omni-layer Intelligent Systems (COINS)*, 2025. arXiv:2502.07255.
- Rikard Laxhammar and Göran Falkman. Sequential conformal anomaly detection in trajectories based on Hausdorff distance. In *Proc. 14th Int. Conf. Information Fusion (FUSION)*, pages 1–8, 2011.
- Rikard Laxhammar and Göran Falkman. Inductive conformal anomaly detection for sequential detection of anomalous sub-trajectories. *Annals of Mathematics and Artificial Intelligence*, 74(1–2):67–94, 2015.
- Jing Lei, Max G’Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.
- Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Schölkopf, Thomas Brox, and Peter Gehler. Towards total recall in industrial anomaly detection. In *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022. arXiv:2106.08265.
- Cheng Shen and Yuewei Liu. Conformal segmentation in industrial surface defect detection with statistical guarantees. *Mathematics*, 13(15):2430, 2025. MDPI; DOI 10.3390/math13152430; arXiv:2504.17721.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Zhiyuan You, Lei Cui, Yujun Shen, Kai Yang, Xin Lu, Yu Zheng, and Xinyi Le. A unified model for multi-class anomaly detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.03687.
- Yang Zou, Jongheon Jeong, Latha Pemula, Dongqing Zhang, and Onkar Dabeer. Spot-the-difference self-supervised pre-training for anomaly detection and segmentation. In *Proc. Eur. Conf. Computer Vision (ECCV)*, 2022. VisA benchmark; arXiv:2207.14315.